Stochastic Strategic Demand–Capacity Balancing Using Model-Free Reinforcement Learning

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01 MOTIVATION

01 Strategic demand-capacity balancing is key to mitigating air traffic inefficiencies

02 Operational uncertainties, particularly weather, challenge effective planning

03 Increasing traffic density and diversity (e.g., UAM) introduce new challenges

04 Traditional decision-support approaches struggle with uncertainty and strategic-tactical integration

02 OBJECTIVES

01 Integrated RL framework

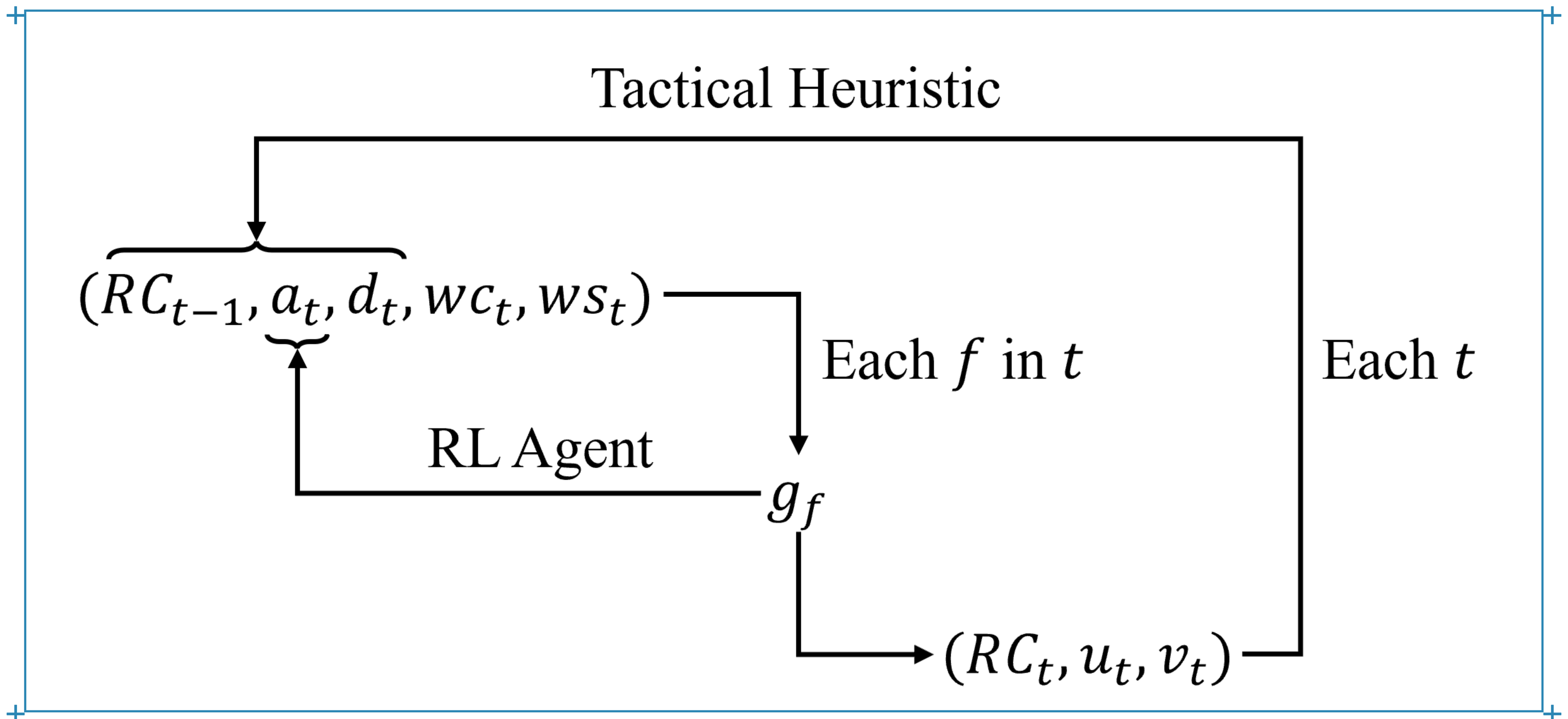
Develop a reinforcement learning framework for demand-capacity balancing that integrates strategic ground-delay allocation with tactical capacity management under uncertain weather forecasts.

02 Two operational paradigms

Demonstrate the applicability of the framework, without modification, to both conventional ATFM at São Paulo/Guarulhos and UAM traffic management in the São Paulo Metropolitan Region.

03 METHOD

One loop, two decision layers — an MDP on 15-minute epochs



Strategic layer — RL agent

For each arrival flight f in epoch t, choose a ground delay g_f in 15-minute increments, up to one hour.

Tactical layer — heuristic

Given the wind state, select a feasible runway configuration (or vertiport side) and set arrival/departure service rates from the capacity envelope.

Coupling

Reallocated demand changes future queues, so the tactical consequence of each delay is fed back into the reward.

STATE

$$s_t = (RC_{t-1}, a_t, d_t, wc_t, ws_t)$$

Previous configuration · arrival queue · departure queue · weather category (VMC / IMC / LIMC) · wind state, encoding which sides stay usable at CW ≤ 20 kt and TW ≤ 5 kt. Horizon T = 96 epochs.

ACTION, MASKED

$$g_f \in \{0, 1, 2, 3, 4\} \times 15 \text{ min}$$

Capped at one hour and at the time remaining in the horizon; a binary mask removes infeasible delays so the agent never explores them.

TACTICAL SERVICE RATES

$$u_t = \min(u_{max}[wc_t, RC_t], a_t)$$
$$v_t = \min(v_{max}[wc_t, RC_t, u_t], d_t)$$

Configuration taken from the operationally valid set $RC_t \in \mathcal{RC}(ws_t)$.

REWARD — THE TACTICAL CONSEQUENCE

$$J^{tac} = \sum_{t=0}^{T} [\alpha(a_t - u_t)^2 + (d_t - v_t)^2 + \delta \cdot \mathbb{1}(RC_t \neq RC_{t-1}) \cdot (\alpha a_t^2 + d_t^2)]$$
$$R_t = -(\Delta J^{tac}(t) + \eta \sum_f g_f), \quad \eta = 0.1$$

Evaluation metric: €45 per delayed departure minute and €90 per airborne arrival minute, following EUROCONTROL (2024).

1. WEATHER DYNAMICS

$$P(METAR_t = j \mid METAR_{t-1} = i)$$

Empirical transition frequencies between VMC, IMC and LIMC at 15-min resolution.

2. FORECAST ERROR

$$P(METAR = j \mid TAF = i)$$

Confusion matrices estimated by comparing forecasts against observed realizations.

TRAINING-TIME WEATHER SAMPLING

$$wc_t \sim P(wc_t \mid wc_{t-1}, \hat{wc}_t) \propto P(wc_t \mid wc_{t-1}) \cdot P(wc_t \mid \hat{wc}_t)$$

The wind state is sampled the same way, after converting wind speed and direction into crosswind and tailwind components relative to runway orientation.

04 CASE STUDIES

Train on forecast, test on observation — 30 test days per case

ALGORITHM

Maskable PPO — Stable Baselines 3 · Gymnasium; 500,000 training episodes per day (conventional), 300,000 (UAM)

DATA

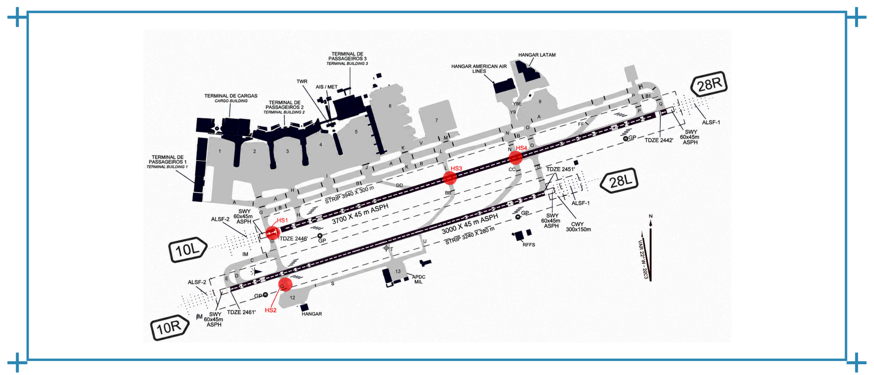
24 months of METAR/TAF (REDEMET) at 15-min resolution; ANAC SIROS schedules corrected with ANAC VRA and FlightRadar24 tracks. TAF for training, METAR for testing.

BASELINE — NO RESCHEDULING

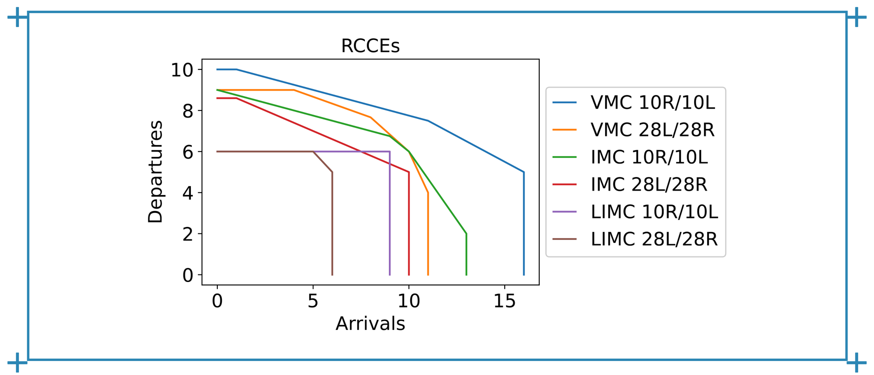
The same tactical heuristic and capacity model with $g_f = 0$ for every flight. The strategic rescheduling cost is omitted in evaluation, so the metric isolates tactical congestion reduction.

CASE 1 — CONVENTIONAL ATM AT SBGR

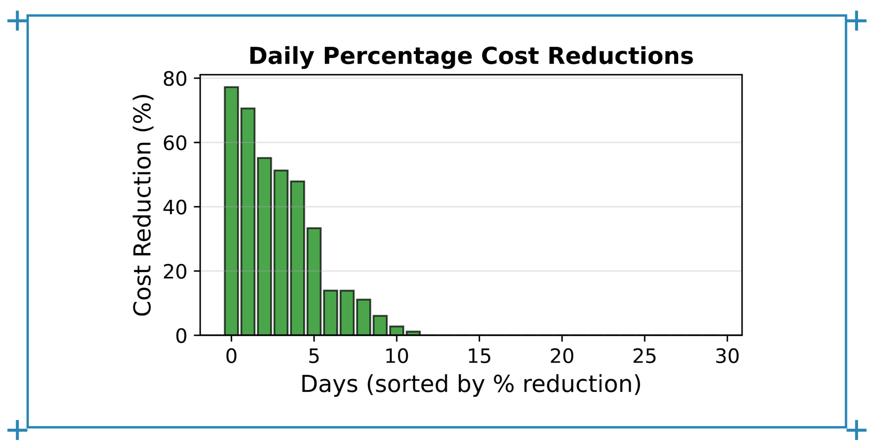
Wilcoxon $p = 0.002183$



300k+ annual operations on two parallel runways; configurations 10R/10L and 28L/28R.



Runway envelopes at SBGR, fitted by quantile regression (Ramanujam & Balakrishnan, 2009): concave and invertible.



Daily percentage cost reduction against the baseline, 30 days.

13.3% aggregate mean-cost reduction

12/30 days improved

0 days degraded

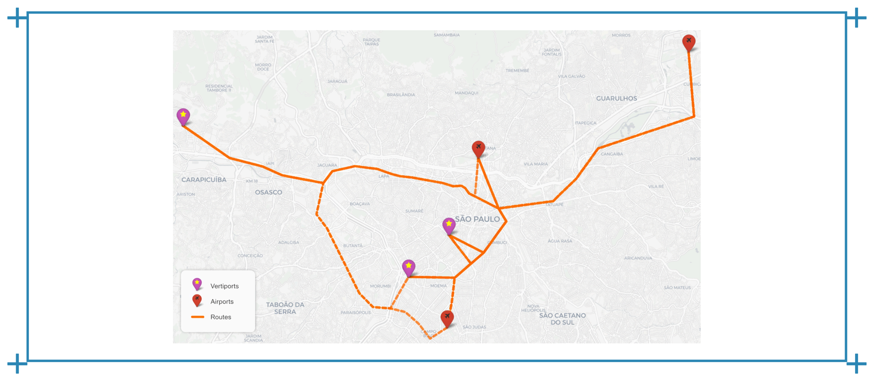
77.2% best single day

1.1% of flights delayed

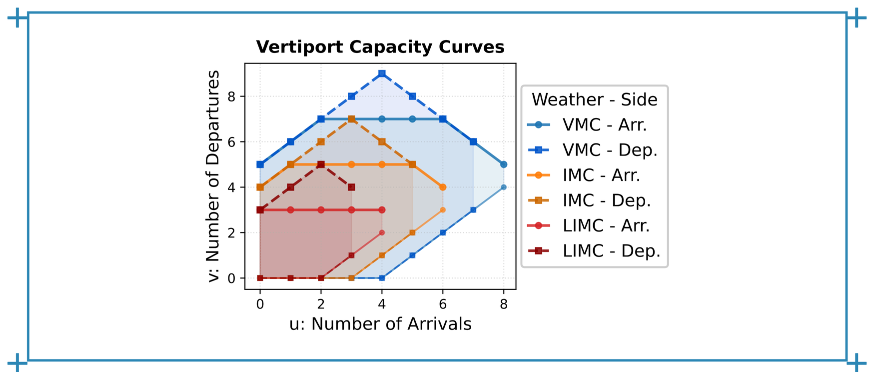
A conservative policy: 98.9% of flights fly as scheduled, and delay is spent only where it removes real tactical congestion.

CASE 2 — UAM TRAFFIC MANAGEMENT, SÃO PAULO METROPOLITAN REGION (RMSP)

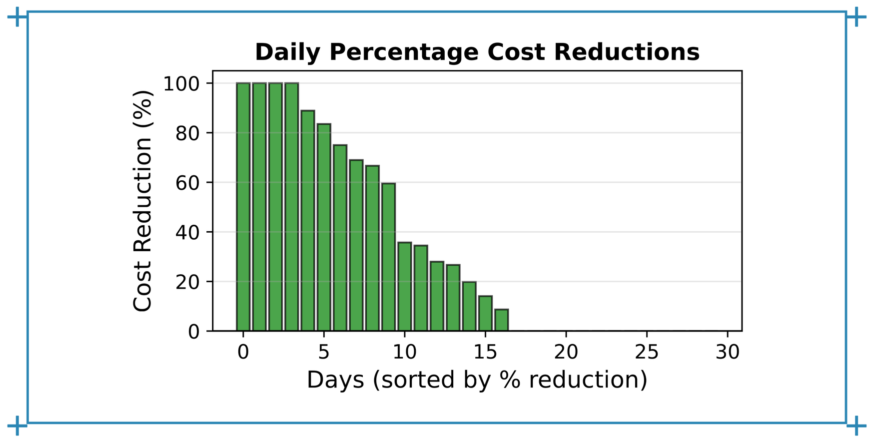
Wilcoxon $p = 0.000281$



Three vertiport candidates in the RMSP, routed to SBGR, SBSP and SBMT over the existing REH corridors.



Vertiport curves. Limited parking imposes a **minimum** departure count per arrival level, so the envelope is non-invertible.



Daily percentage cost reduction against the baseline, 30 days.

64.7% aggregate mean-cost reduction

23/30 days improved

83.5% worst-case cost cut

43.9% mean daily reduction

4.4% of flights delayed

The synthetic UAM baseline is deliberately not pre-optimized against vertiport capacity, leaving more headroom than an operationally planned schedule.

GROUND DELAY ALLOCATED — 10,144 / 2,857 FLIGHTS

DELAY	SBGR	UAM
0 slots	98.9%	95.6%
1 slot (15 min)	0.7%	2.2%
2 slots (30 min)	0.4%	1.8%
3 slots (45 min)	0.0%	0.2%
4 slots (60 min)	0.0%	0.3%

05 FUTURE PERSPECTIVES

What the framework contributes, and what it does not yet cover

CONTRIBUTIONS

01 Model-free RL framework for strategic DCB with probabilistic weather representation

02 Integrated strategic and tactical optimization of delay, configuration and service rates

03 Action masking that removes infeasible ground-delay exploration and accelerates learning

04 Demonstrated transfer across conventional ATM and UAM with the same algorithm

BEYOND A SINGLE AIRPORT

Network effects, airline recovery, passenger connections and multi-airport coordination remain unexplored, as do wind-state definitions at other aerodromes.

WEATHER REPRESENTATION

Route-level low-altitude weather and finer variables would improve realism over VMC/IMC/LIMC categories derived from airport METAR and TAF.

Selected references: Richetta & Odoni (1994); Ramanujam & Balakrishnan (2009); Bertsimas et al. (2011); Vascik & Hansman (2019); Estes & Ball (2020); Shone et al. (2021); Jones et al. (2023); EUROCONTROL (2024); Carvalho & Murça (2025).

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orbolato.github.io/dasc2026-paper-276

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